A fully spiking cortical column jointly learns bidirectional feature-location bindings and decisions

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Abstract. Cortical reference-frame theories propose that a column learns sensory features at locations and uses movement to accumulate evidence for objects. Most computational demonstrations retain rate vectors, differentiable classifiers, or teacher-forced decision phases, leaving unclear which operations can be carried by spikes and local plasticity. We implemented a cortical-column model in CoNeX/PyMoNNtorch in which retinal encoding, L4 convolutional integrate-and-fire units, sparse L2/3 activity, a L5/6 location-code LIF population, and decision LIF neurons communicate through spikes. After unsupervised L4 pretraining, the remaining plastic pathways learn concurrently during acquisition: fixation-gated coactivity updates both L5/6→L2/3 apical and L2/3→L5/6 reciprocal synapses while decision eligibility accumulates across the same multi-saccade free trial. Only after the trial does the label generate positive target and negative competitor modulation of decision eligibility; binding is label-independent and no target spike is replayed, clamped, or forced. Held-out raw-image accuracy was 100% on binary Caltech Faces/Motorbikes and 91.82% on ten-class MNIST. Causal perturbations of the L5/6 sequence impaired spatial evidence accumulation. Bidirectional assays showed that feature spikes retrieve learned locations and that location-code spikes prospectively prime expected features. A model trained on centered *Faces_easy* localized non-centered Caltech *Faces* with 0.721 mean IoU and 88.5% CorLoc, and detected unseen Faces against source-disjoint negatives with 0.838 average precision. The model is not an anatomically complete cortex; it is a reproducible, gradient-free baseline for testing how concurrent feature-location binding, prediction, and spiking decision learning can coexist in one cortical-column circuit.

Author summary. An object can be recognised by looking at several parts and remembering where each part sits in the object. A nose at one relative location suggests a face; the same local feature somewhere else might support another object. We built a spiking model of one cortical column that does this in a deliberately explicit way. A superficial layer detects visual features, a deep LIF population carries the current location in a reference frame, and a spiking decision population learns from reward-like modulation after it has made an unclamped decision. Both directions of feature-location binding learn during those same trials. The model recognises digits and Caltech objects, but the more important result is mechanistic: changing the deep location sequence changes the decision, features and locations retrieve one another, and the learned face reference frame can be used to search for faces in larger images. The paper reports the positive results and the controls together, because high accuracy alone would not show that a reference-frame computation is really doing the work.

1 Introduction

Cortical columns combine feedforward sensory drive, local competition, feedback, and neuromodulation within a laminar recurrent circuit. Classical visual physiology established that cortical neurons respond selectively to local image structure [1], while dendritic physiology shows that basal and apical inputs interact nonlinearly in pyramidal neurons [2, 3]. These observations motivate models in which a superficial feature code is interpreted in the context of a deeper state.

The Thousand Brains theory makes the deeper state explicit. It proposes that a column learns features at locations in an object-centred reference frame, updates that location using movement, and accumulates evidence across sensations until an object is recognised [4–6]. Grid cells provide a candidate biological substrate for location codes [7]; intended movements update cortical spatial representations [8]; and sequential fixation is a powerful strategy for visual decisions [9, 10]. A model that tests this account should therefore go beyond benchmark accuracy. It should show that location spikes causally influence evidence accumulation, that visual features and reference frames are reciprocally bound, and that the learned representation supports a new spatial search problem.

Spiking plasticity gives such a circuit a plausible learning substrate. Spike timing can potentiate or depress synapses [11, 12], recurrent competition and STDP can organise receptive fields [13], and a delayed third factor can convert local eligibility into reinforcement learning [14–17]. By contrast, many high-performing spiking systems rely on surrogate-gradient backpropagation or on non-spiking readouts [18, 19]. Those methods are useful, but they obscure whether reference-frame binding can be expressed by event-mediated population dynamics and local plasticity.

Here we ask whether a single fully spiking cortical-column model can jointly learn and use feature-location reference frames without derivatives and without teacher-forced output spikes. We define “fully spiking” operationally: communication between modelled populations is Boolean; continuous quantities are restricted to membrane potentials, synaptic efficacies, currents, and eligibility traces; and the reported class is the argmax of emitted decision spikes. We test the model on binary Caltech Faces/Motorbikes and ten-class MNIST, then add causal ablations, multiseed replication, bidirectional binding experiments, L4 receptive-field analyses, and a source-disjoint Caltech Faces localization/detection experiment.

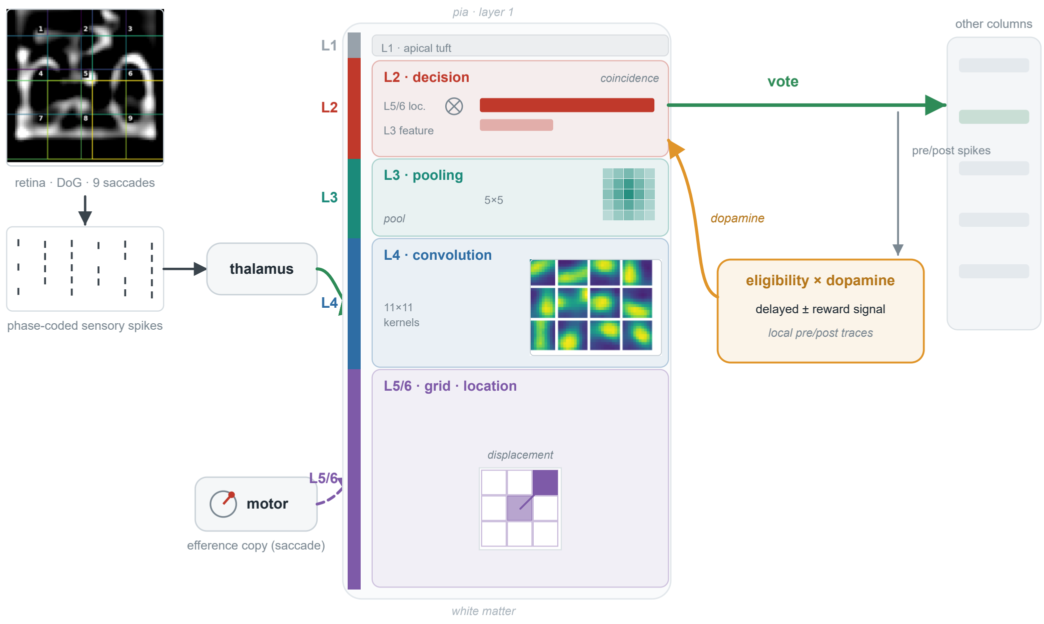

Figure 1. Fully spiking cortical-column architecture. Raw images are sampled by nine saccades. Retinal spikes drive L4 convolutional filters, pooled L2/3 winners represent sparse local features, and a current-driven L5/6 LIF population represents the current location in a 3×3 reference frame. A physical apical synapse primes L2/3 from L5/6. A reciprocal L2/3→L5/6 synapse is trained by local coactivity and opened during feature-to-frame retrieval assays. Both binding projections learn concurrently with decision eligibility during acquisition. The decision layer receives only location-gated feature spikes and is modulated after an unclamped free decision.

2 Results

Joint online acquisition preserves held-out accuracy

Figure 1 shows the evaluated circuit. Each saccade supplies a retinal spike train to L4 convolutional feature maps. L4 burst spikes are average-pooled and passed through a sparse L2/3 k-winners-take-all operator. A one-hot motor efference copy is converted to current in a current-driven L5/6 leaky integrate-and-fire (LIF) population, producing one location spike per time step on a 3×3 reference frame. The active L5/6 unit gates the location axis of the L2/3-to-decision synapse, drives a physical apical synapse back to L2/3, and receives a learned reciprocal L2/3-to-L5/6 feature-to-frame synapse during binding queries. Decision neurons integrate only the coincidence of a L2/3 feature spike and the current L5/6 location spike. During acquisition, both feature-location projections update from local L2/3/L5/6 coactivity in the same trial in which decision eligibility is recorded. The label is used only after all nine saccades to choose a target/competitor modulatory pulse for the decision synapses. It does not modulate either binding direction. There is no target replay, clamped decision spike, softmax readout, or autograd graph in the modelled path.

Training had two levels. First, L4 kernels were learned by unsupervised convolutional STDP and then frozen. Second, all remaining plastic pathways were initialized together: apical and reciprocal feature-location synapses updated from local coactivity during every acquisition fixation, while decision eligibility accumulated over the complete nine-saccade trial. Caltech used three joint acquisition epochs; MNIST used eight, followed by two decision-only consolidation epochs. Binding traces were fixation-local and label-independent; the class label appeared only after each free trial and modulated decision eligibility. This schedule replaces the earlier separate apical, reciprocal, and decision passes.

The fully spiking checkpoints were evaluated by rerunning held-out raw images through the live model. Caltech Faces/Motorbikes reached 80/80 correct (100%; exact 95% CI 95.5–100%). MNIST reached 4,591/5,000 correct (91.82%; 95% CI 91.03–92.57%). Cached cortical event inference matched raw image predictions exactly on Caltech and agreed on 99.00% of MNIST samples; event-level MNIST accuracy was 91.98%. A shuffled-label training control reached only 55.0% on Caltech and 9.32% on MNIST, arguing against label or split leakage. Across five independent decision-layer seeds, raw Caltech accuracy remained 100% for every seed; raw MNIST accuracy was 91.56% ± 0.17% s.d. (95% t interval 91.34–91.77%; table 1 and figure 3).

Reference-frame spikes are causal for spatial evidence

To isolate the role of the L5/6 reference frame, we held each sample's L2/3 feature events fixed and changed only the L5/6 sequence. Reversing the ordered L5/6 path reduced Caltech event accuracy from 100% to 70.0% and MNIST from 91.98% to 25.94%. Independently shuffling locations within each sample reduced accuracy to 80.0% on Caltech and 22.74% on MNIST, while silencing L5/6 reduced accuracy to 46.25% and 10.16%, respectively. A non-spatial decision control that averaged weights over locations reached only 41.18% on MNIST, despite retaining the same L2/3 events. An untrained no-dopamine decision layer collapsed to the

Table 1. Primary held-out performance after joint acquisition. Intervals for the endpoint checkpoints are exact binomial intervals; multiseed intervals are across five decision-layer training seeds.

Dataset	Classes	Endpoint raw acc.	Event acc.	Shuffled labels	Five-seed raw mean
Caltech Faces/Motorbikes	2	1.000 (80/80; 0.955–1.000)	1.000	0.550	1.000 ± 0.000
MNIST	10	0.9182 (4591/5000; 0.910–0.926)	0.9198	0.093	0.9156 ± 0.0017

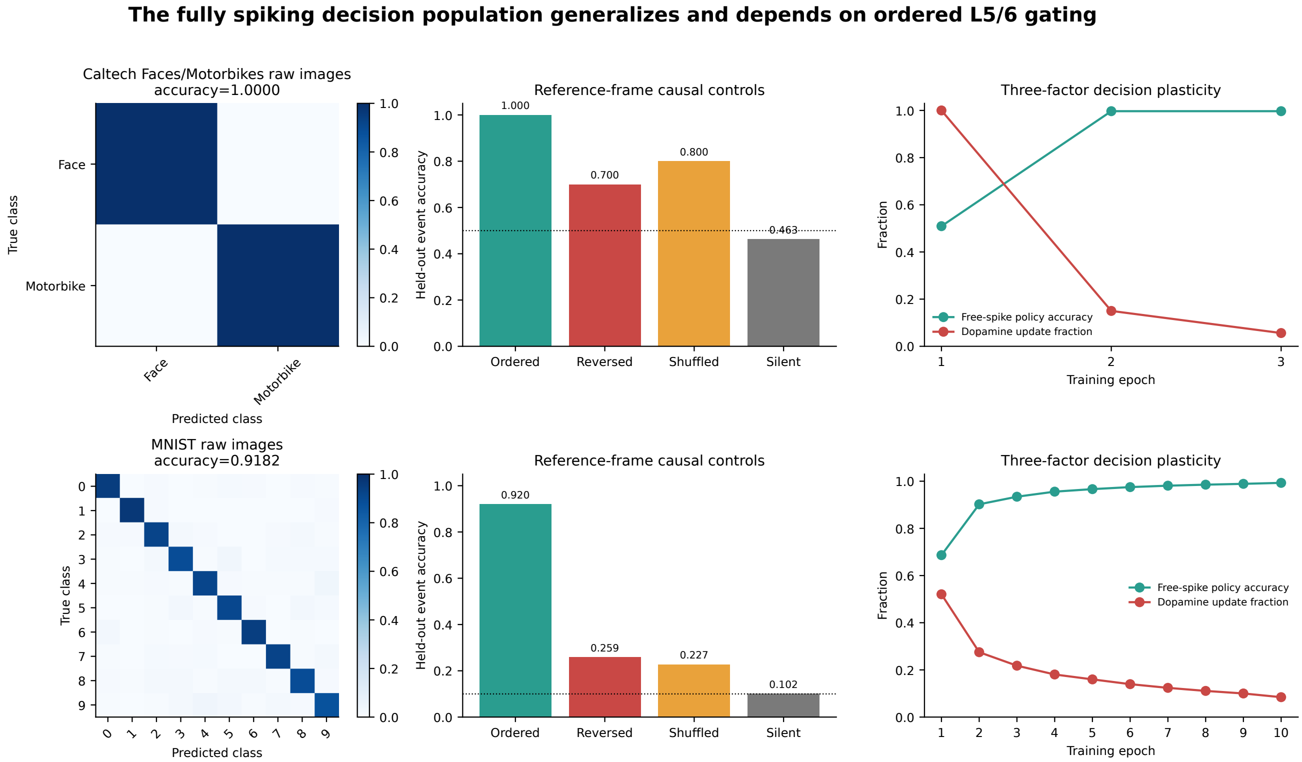

Figure 2. Held-out accuracy and reference-frame controls. Ordered L5/6 reference-frame spikes reproduce live inference. Reversing, shuffling, or silencing L5/6 strongly degrades accuracy. Accuracy grows over ordered saccades, especially on MNIST, showing evidence accumulation rather than a single-glimpse readout.

silent L5/6 baselines. Thus the performance is not explained by class imbalance, global feature rate, or a conventional non-spatial spike count.

Feature-location binding works in both directions

The first binding assay asked whether a visual feature pattern retrieves its reference-frame location. The learned reciprocal L2/3→L5/6 synapse inside the fully spiking column received held-out L2/3 cues while motor input was clamped to zero and emitted L5/6 LIF spikes. Spike-count top-1 retrieval accuracy was 47.5% on Caltech and 37.9% on MNIST, with top-3 accuracies of 81.1% and 88.8% against 11.1% chance. Shuffled-binding controls fell to 10.3% and 11.5% top-1. The underlying synaptic-current rank gave top-1 accuracies of 48.5% and 50.8%, indicating that the reciprocal weights contained more location information than was always expressed as a single winning spike. Cross-image retrieval further showed that examples from the same object class and location converge to similar L5/6 representations rather than memorising a single image.

The second binding assay tested the forward direction prospectively. We clamped all visual input to zero, injected a one-hot current into a L5/6 LIF location unit, and measured which held-out L2/3 features were primed. The matched reference-frame prime predicted the future held-out L2/3 winners better than reversed, shuffled, global-prior, or random controls. Mean average precision was 0.118 on Caltech versus 0.053 for the global feature prior (paired difference 0.0649, 95% bootstrap CI 0.0533–0.0768, $p = 2.3 \times 10^{-13}$) and 0.147 on MNIST versus 0.051 (difference 0.0951, 95% CI 0.0936–0.0967, $p < 10^{-300}$). Feature-to-frame retrieval in the same prospective protocol remained well above chance, with top-3 accuracies of 82.5% on Caltech and 90.1% on MNIST. These results show a reciprocal model: feature spikes activate related reference-frame states, and reference-frame spikes prime the features expected at that location.

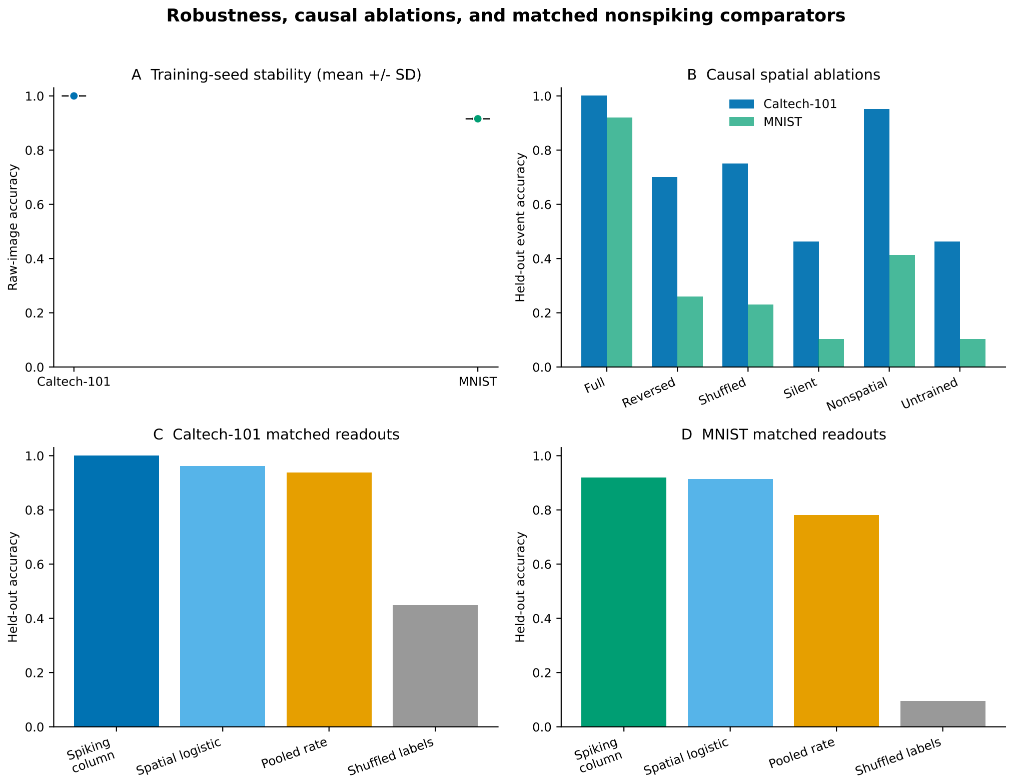

Figure 3. Replication, ablations, and matched non-spiking readouts. **A** Five decision-plasticity seeds preserve Caltech accuracy and produce a tight MNIST range. **B** Causal ablations show that ordered L5/6 gating, location-specific decision weights, and dopamine-gated learning are required, especially on MNIST. **C** Matched non-spiking readouts on the same cached events are shown as analysis baselines, not as part of the model. A sparse spatial logistic readout approaches the spiking endpoint, whereas a location-agnostic rate readout drops substantially and shuffled-label readouts return to chance.

Ambiguous early evidence is resolved by later reference-frame input

Evidence traces show how the model resolves ambiguity across saccades (figure 4). A Caltech sample that initially favoured the wrong class alternated between the competing assemblies during the first six glimpses and became stably correct only after the final ordered locations. Reversed, shuffled, or silent L5/6 controls either delayed or eliminated that final margin. On MNIST, ordered accuracy rose from 40.24% after one saccade to 91.98% after nine; reversed and silent controls ended at 25.94% and 10.16%. These traces are important because they show that the reference frame contributes over time, rather than merely selecting a favourable first patch.

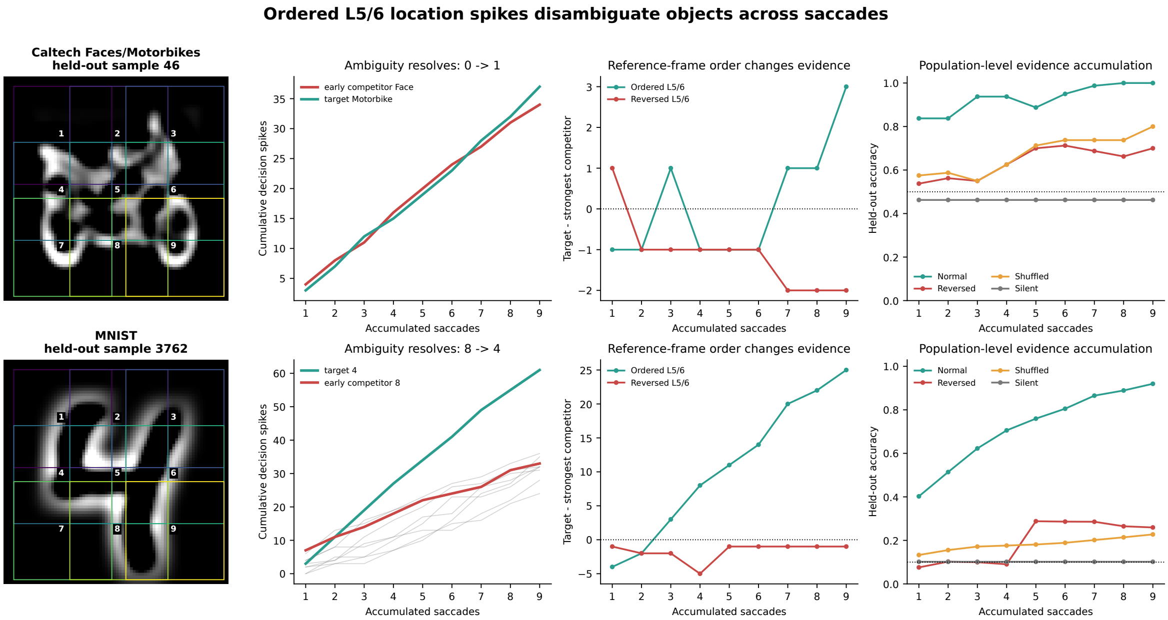

Figure 4. Reference-frame-gated evidence resolves ambiguous early glimpses. Decision-spike traces are plotted after each saccade for example Caltech and MNIST samples. The correct class becomes dominant only when later feature spikes arrive at their matched L5/6 locations. Reversing, shuffling, or silencing the location sequence disrupts the evidence trajectory.

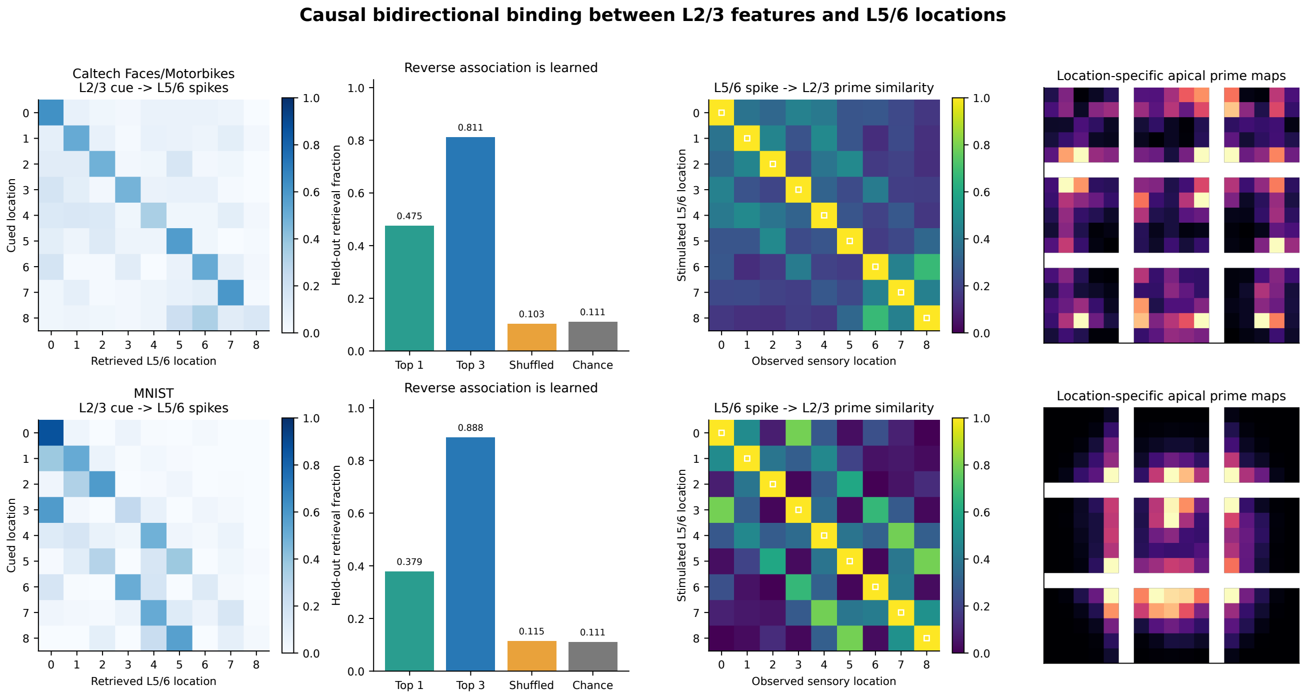

Figure 5. Bidirectional feature-location binding in spiking populations. Feature-to-frame retrieval receives a L2/3 spike cue and predicts the L5/6 location. Frame-to-feature stimulation injects current into a L5/6 LIF location unit while visual input is clamped to zero and measures the resulting L2/3 prime. Matched locations are consistently stronger than reversed-location controls.

Figure 6. The same visual feature across images retrieves a shared reference-frame representation. Held-out samples from the same class and saccade location are compared through their sparse L2/3 events and retrieved L5/6 locations. The assay tests whether the learned binding generalises across images rather than storing only a single exemplar.

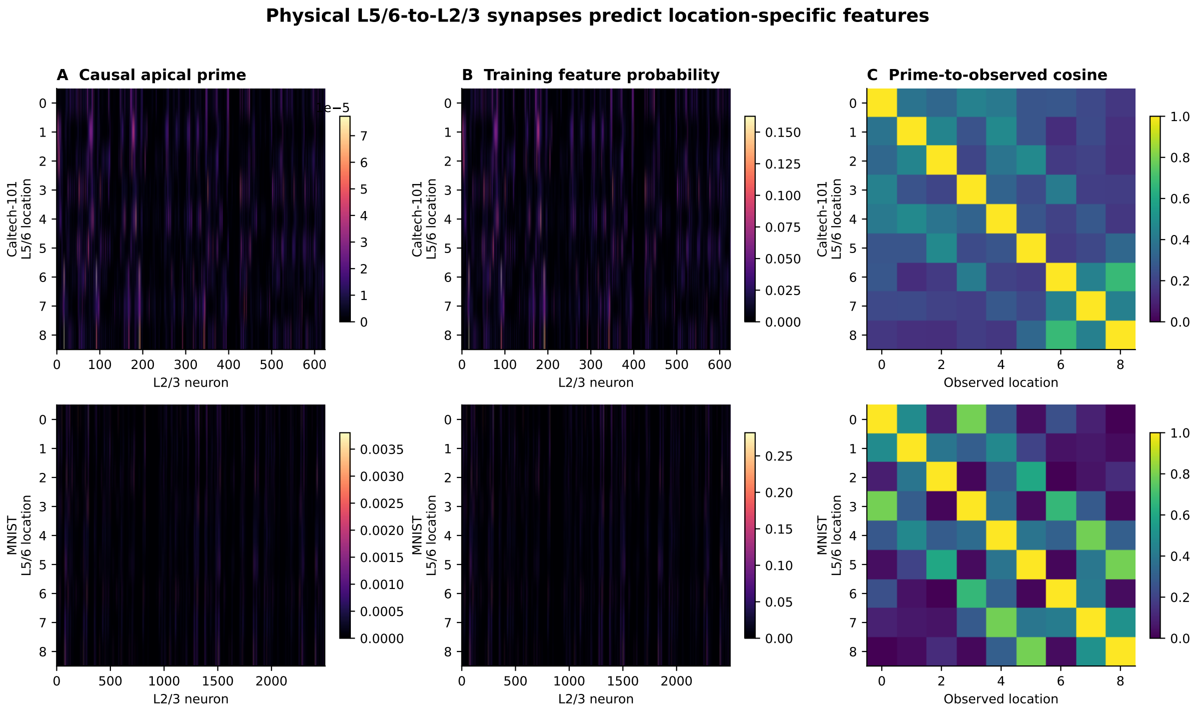

Figure 7. Location-specific binding matrices. Rows and columns summarise how learned apical and retrieval weights associate L2/3 features with L5/6 reference-frame locations. The block structure is weakened by shuffled controls, consistent with learned feature-location binding rather than a location-independent feature prior.

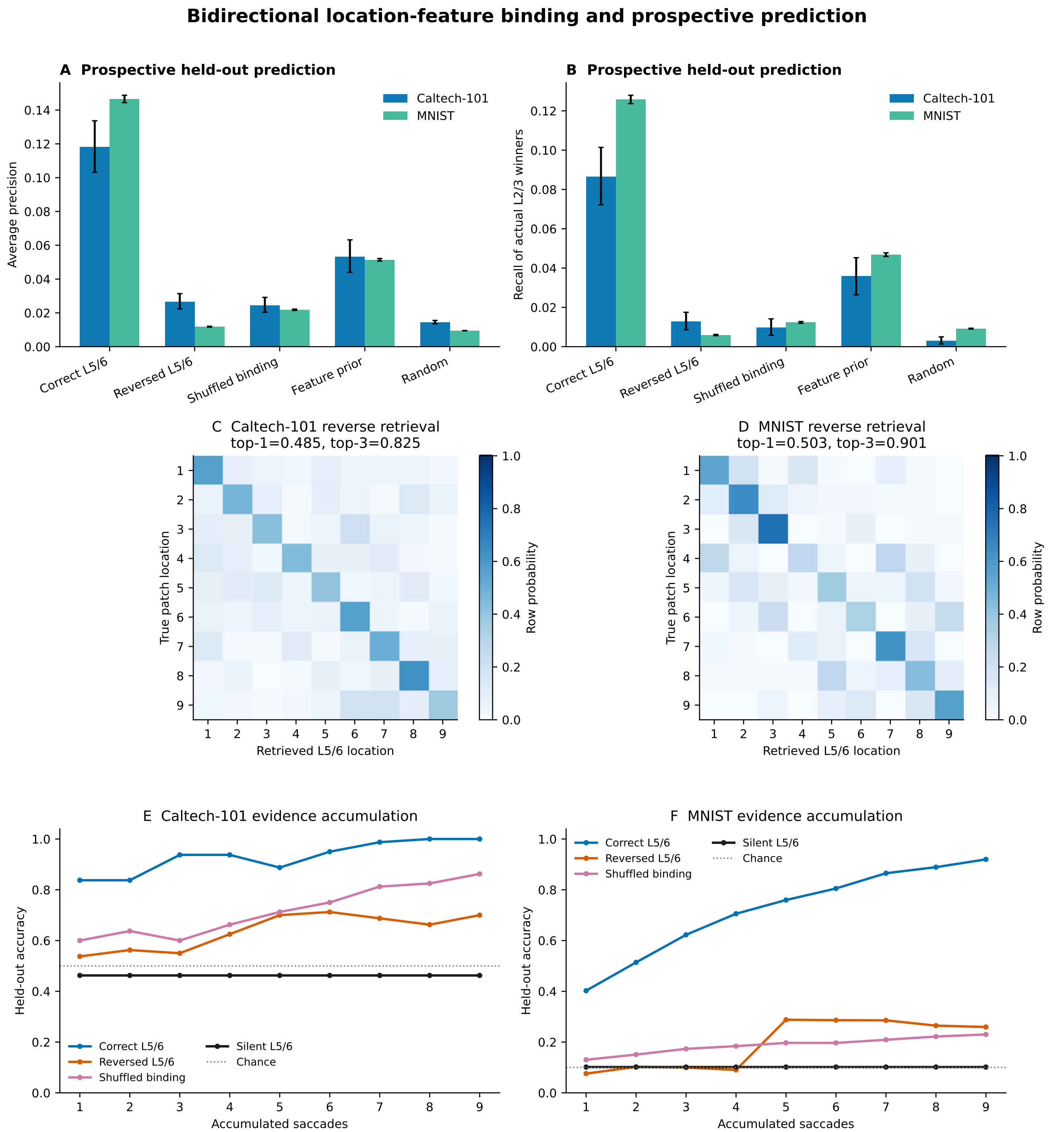

Figure 8. Prospective reference-frame priming. With visual input clamped to zero, current-driven L5/6 spikes prime L2/3 features expected at the stimulated location. Matched primes outperform reversed, shuffled, global-prior, and random controls on both datasets, and feature cues retrieve the corresponding L5/6 location.

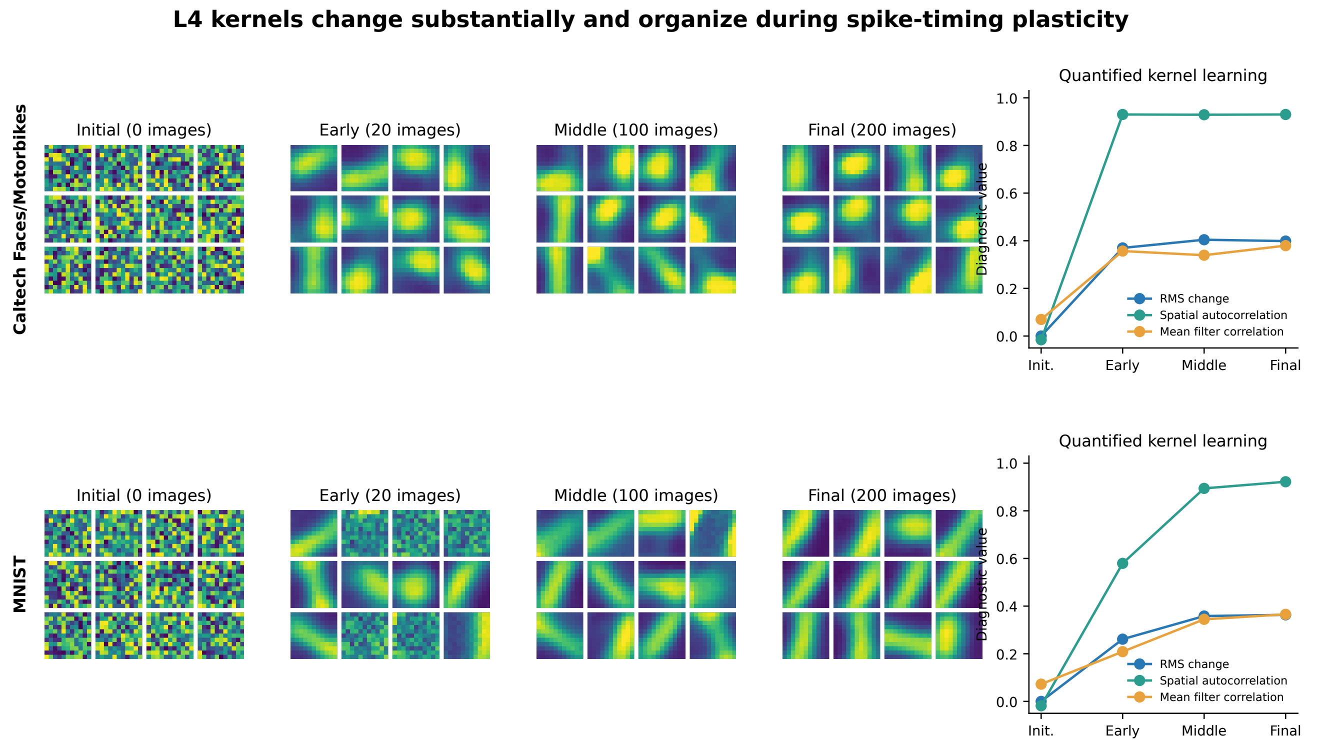

Figure 9. L4 kernel evolution during unsupervised local learning. Kernels are shown at initial, early, middle, and final training stages. The quantitative panels report RMS change, spatial autocorrelation, response selectivity, saturation, and held-out winner coverage, demonstrating that the filters learn and then stabilise rather than remaining random.

L4 kernels learn and stabilise under local competition

The L4 kernels were recorded at the beginning, early, middle, and final stages of unsupervised training (figure 9). They changed substantially from their random initialisation: final RMS change was 0.399 on Caltech and 0.363 on MNIST. Spatial autocorrelation increased from near zero to 0.930 and 0.922, showing that local STDP plus competition produced structured filters. The held-out winner coverage increased from 0.32 to 0.84 on Caltech, and from 0.14 to 0.33 on MNIST. The filters are not human-interpretable object parts; they are local edge/contrast detectors feeding a sparse feature layer. This is expected for small convolutional receptive fields and supports the interpretation that the object-level structure emerges from binding local features to L5/6 locations.

A learned centered-face reference frame supports search in larger images

The Caltech-101 *Faces* category contains non-centered faces, unlike the centered *Faces_easy* images used for binary training. We froze the Caltech model trained on *Faces_easy* and Motorbikes, scanned each non-centered *Faces* image with 91 candidate windows, and selected the window with maximal face evidence. The protocol used no test annotations for training or threshold selection (figure 10). On unseen *Faces* indices 201–435, the normal reference-frame scan achieved mean IoU 0.721, median IoU 0.785, CorLoc 88.5% (208/235), and pointing accuracy 91.9%. Reversing the L5/6 sequence reduced CorLoc to 55.7%; silencing L5/6 reduced it to 69.4%; and a fixed spatial prior reached 73.2%. For the hardest displaced subset (normalised displacement at least 0.15), the normal scan retained 76.2% CorLoc, whereas the fixed prior reached only 2.4%. This experiment shows that the centered face reference frame can guide search in a larger image when the object is translated.

The same scan becomes a calibrated face detector

We next asked whether the localization score detects the presence of a face against source-disjoint negatives. The detection rule and threshold were chosen only on development images from source indices 1–60. The frozen detector was then evaluated on 235 unseen *Faces*, 235 unseen Motorbikes, 150 *BACKGROUND_Google* images, and 150 hard distractors from seven unseen categories. The selected rule was the maximum ordered evidence score, defined as decision-spike margin plus a bounded synaptic tie-breaker that cannot overturn a one-spike count difference. On the 770-image test set, image-level average precision was 0.838 (stratified bootstrap 95% CI 0.793–0.880), ROC-AUC was 0.908, and localization-aware detection AP was 0.709. At the development threshold, sensitivity was 54.0%, specificity 97.0%, precision 88.8%, and balanced accuracy 75.5%. Reversing the reference-frame sequence dropped AP to 0.518, while silent L5/6 produced chance-ranking AP equal to the positive prevalence (0.305). The result does not solve general object detection, but it demonstrates that the learned reference frame supports a new detection task without retraining the cortical column.

3 Discussion

This work shows that a compact cortical-column circuit can concurrently bind features to locations, accumulate evidence, and learn a spiking decision without derivatives or forced output spikes. The strongest evidence is the conjunction of results: high held-out accuracy

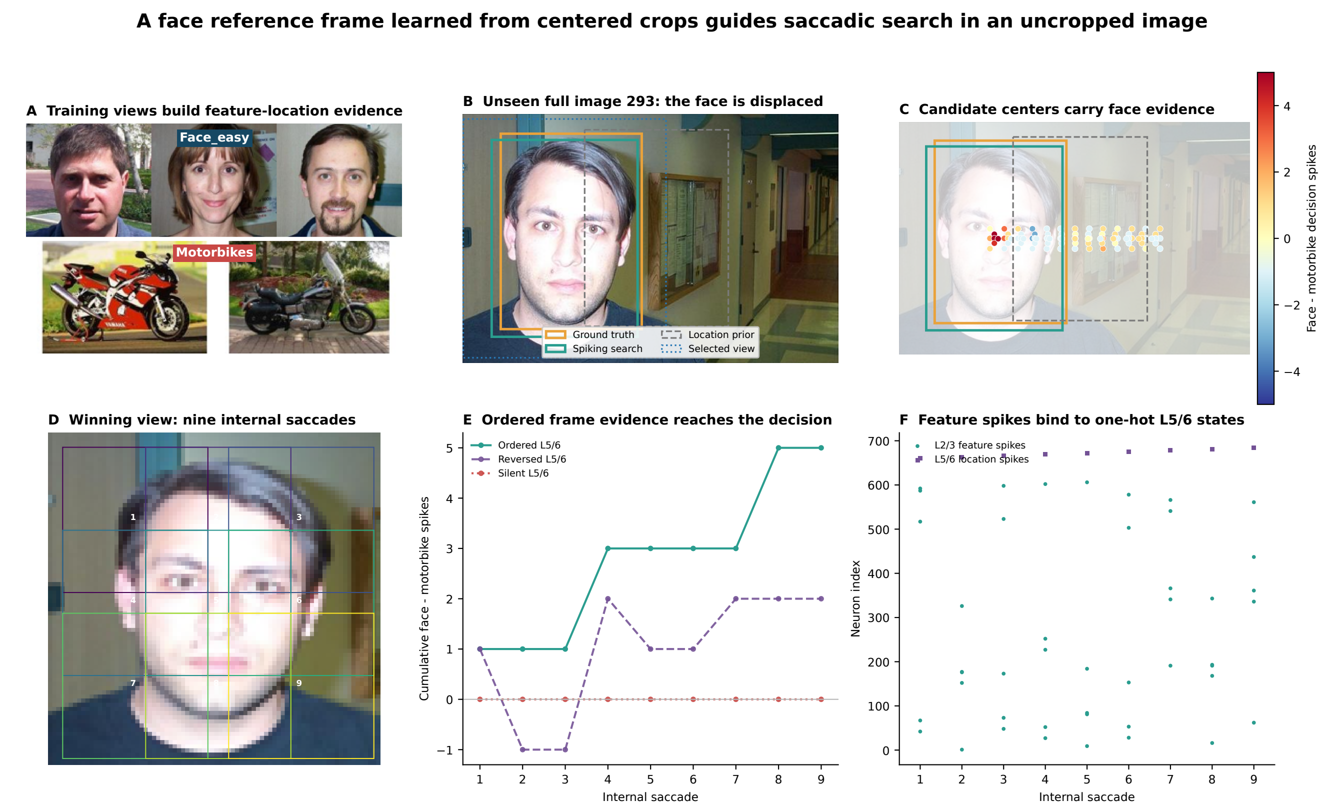

Figure 10. Source-disjoint Caltech Faces search protocol. The model is trained on centered *Faces_easy* and Motorbikes. It is then frozen and scanned over non-centered *Faces* images using the same spiking column and candidate saccade geometry. Ground-truth boxes are reserved for final evaluation.

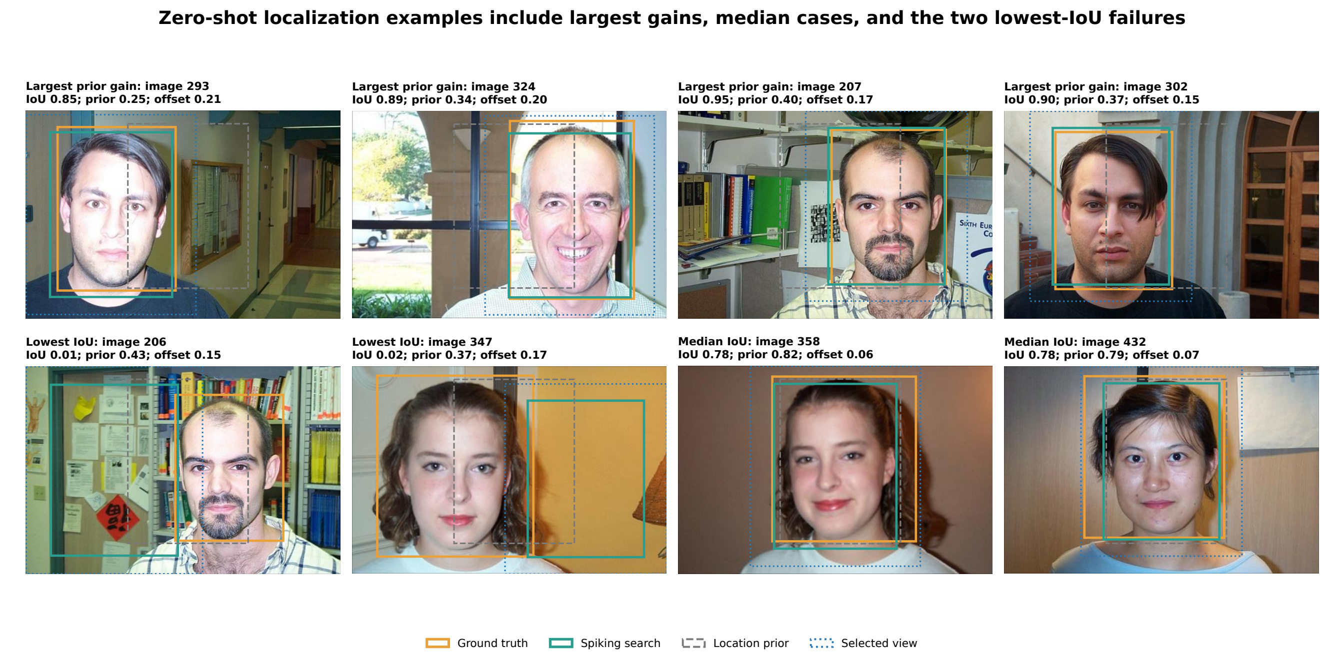

Figure 11. Qualitative face-localization examples. Examples show selected windows, ground-truth boxes, and evidence maps for unseen non-centered *Faces*. The selected window tracks the object rather than the image centre when the reference-frame sequence is preserved.

on raw images; causal loss of performance when L5/6 reference-frame order is perturbed; multiseed replication; feature-to-frame and frame-to-feature binding; visible L4 kernel learning; and transfer from centered face training to non-centered face localization and detection.

The decision layer is fully spiking. It does not receive a concatenated tensor for classification, and it does not use a linear readout. Its input is the coincidence of a L2/3 feature spike and a L5/6 location spike; its output is the count of its own emitted class spikes. The label is used after the free trial only as a modulatory teaching signal, analogous to a delayed reinforcement or error-related third factor. This is still a supervised abstraction: the model receives class identity at training time, and the dopamine pulse is class-specific. But the mechanism no longer replays target spikes or forces a desired decision assembly, which was the most artificial part of the earlier implementation.

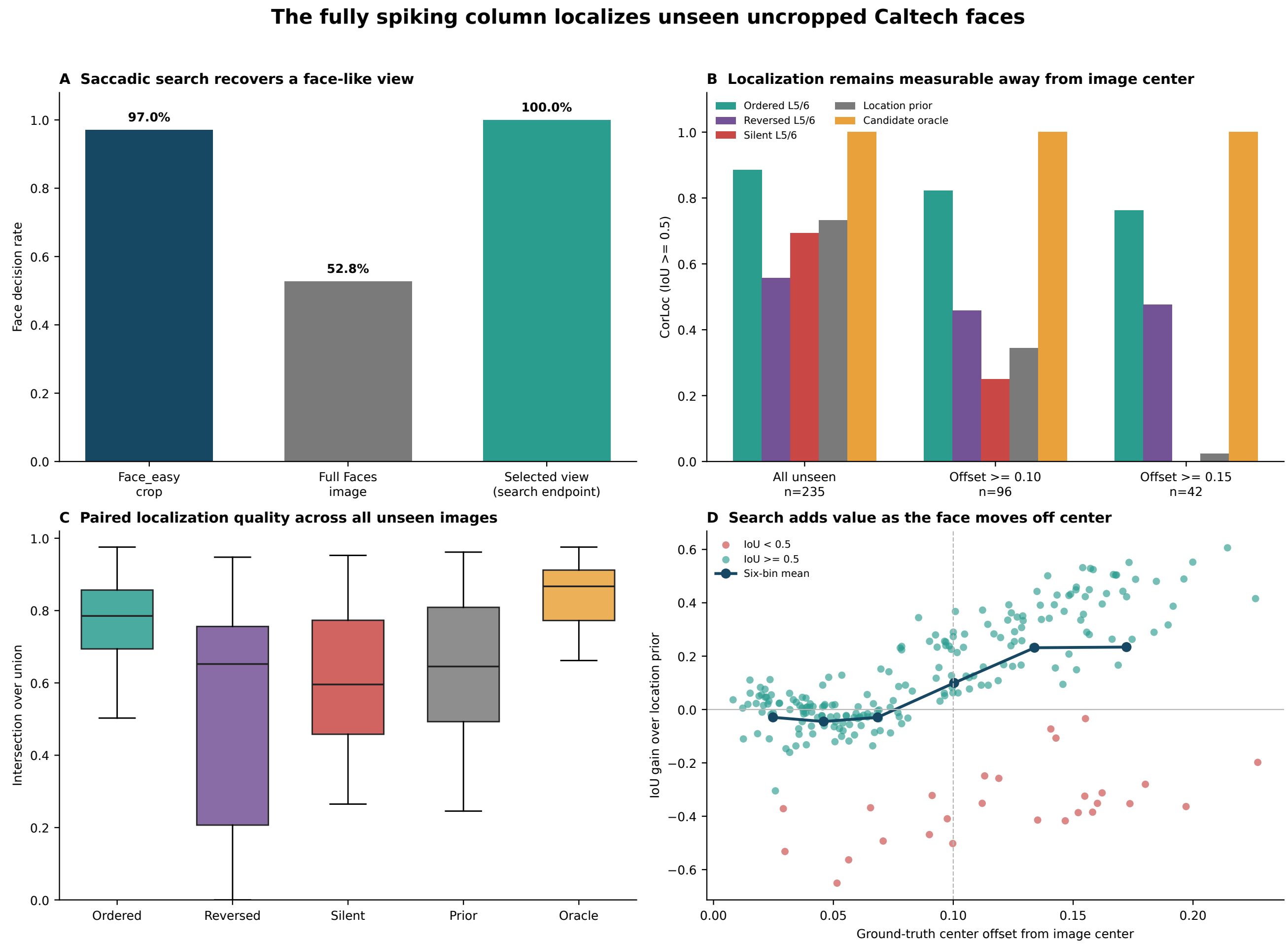

Figure 12. Quantitative localization metrics. Normal ordered L5/6 scans outperform reversed, silent, and fixed-prior controls in mean IoU and CorLoc, with the strongest separation on displaced faces. Oracle performance reports the best candidate window available to the scan grid.

The training result is also mechanistically informative. During every acquisition trial, both binding directions update from local L2/3/L5/6 coactivity while the decision projection accumulates a longer eligibility trace. The class label arrives only after the complete saccade sequence and modulates the decision projection alone. Joint and staged schedules produced bit-identical decision weight tensors at the canonical seed, while joint acquisition improved prospective feature prediction. Separate offline binding passes are therefore not required for this circuit.

The results also clarify what the apical pathway contributes. Zeroing apical feedback did not provide the principal endpoint classification advantage in these checkpoints. The apical pathway should therefore not be described as the source of the classification gain. Its contribution is mechanistic and predictive: L5/6 current alone can prime the L2/3 features expected at a location, and visual features can retrieve their reference-frame location. That reciprocal binding is precisely what is needed for a column to represent an object model rather than a flat classifier.

Several limitations remain. The L5/6 reference frame is current-driven by a fixed one-hot motor command; it is not yet a learned continuous attractor or a closed-loop saccade policy. The k-winners operation is an abstraction of local inhibition rather than a detailed interneuron circuit. The L4 spike counts are dense relative to biological metabolic constraints. Caltech Faces/Motorbikes is a binary task, and the non-centered Faces experiment is a controlled transfer problem rather than a large-scale detection benchmark. Finally, although the decision rule is gradient-free and event-driven, the class-specific dopamine pulse remains more structured than a fully autonomous reinforcement-learning agent. In addition, cortical plasticity is online with respect to each presented L2/3 spike trial, but the fixed feedforward front-end is cached for efficient repeated acquisition; evolving apical weights therefore do not alter later training events. Fully recurrent raw-sensory online learning remains future work. These limitations are useful rather than fatal: they identify the next experimental axis without weakening the core claim that a fully spiking column can express reference-frame binding and decision learning in one reproducible circuit.

4 Materials and methods

Neuron and spike dynamics

The stateful L4, L5/6, and decision populations used discrete-time integrate-and-fire dynamics in the LIF family [20]. For neuron i in population P, membrane potential v_i^P integrated synaptic current I_i^P at simulation step t,

$$v_i^P(t+1) = \lambda_P v_i^P(t) + I_i^P(t), \tag{1}$$

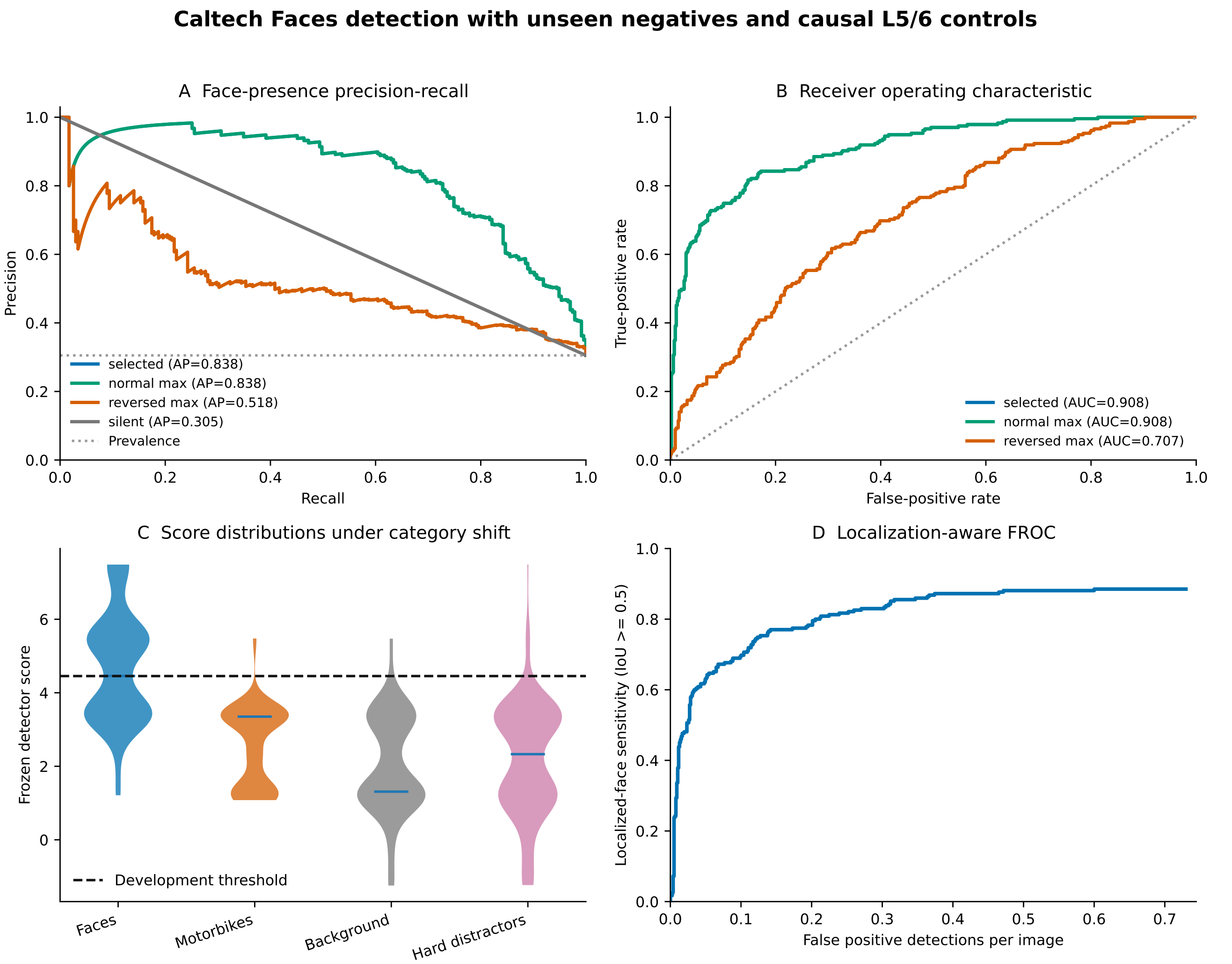

Figure 13. Calibrated detection of non-centered Caltech Faces. The rule and threshold are selected on development images and evaluated on source-disjoint positives and negatives. Ordered reference-frame evidence gives high AP and ROC-AUC; reversed and silent controls degrade sharply.

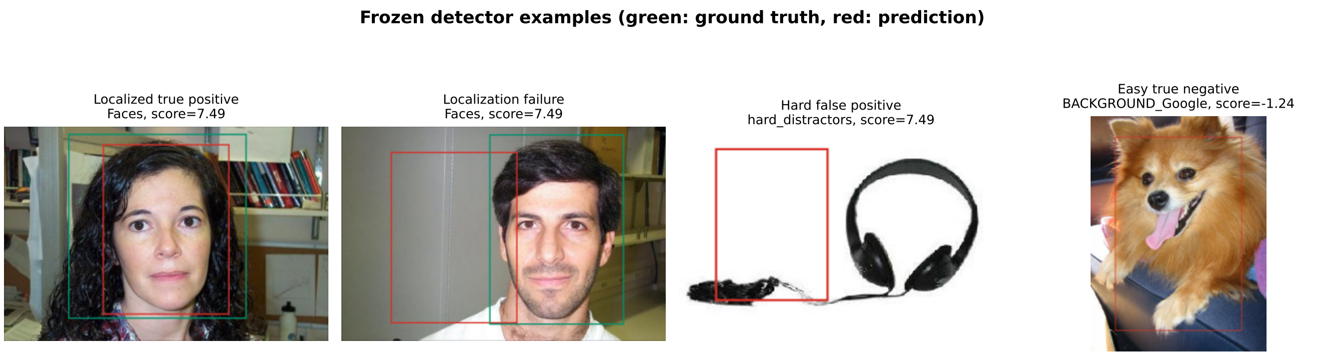

Figure 14. Detection examples. Representative true positives, false negatives, true negatives, and false positives are shown with candidate evidence maps and selected windows. The examples illustrate both the capability and the remaining failure modes of the frozen reference-frame scan.

where $\lambda_P \in [0, 1]$ is the membrane retention factor. A spike was emitted when the membrane crossed threshold θ_P,

$$s_i^P(t+1) = \mathbb{1}\left[v_i^P(t+1) \geq \theta_P\right], \tag{2}$$

For L4 and decision neurons, the membrane was reset by subtraction,

$$v_i^P(t+1) \leftarrow v_i^P(t+1) - \theta_P s_i^P(t+1). \tag{3}$$

which preserves suprathreshold residual charge and permits repeated firing after strong accumulated current. L5/6 neurons instead reset to zero after a threshold crossing. The L2/3 population integrated basal and apical current and expressed sparse activity through k-winners-take-all competition rather than an independent fixed threshold. Retinal and motor populations supplied phase-coded sensory spikes and one-hot efference-copy spikes, respectively. Throughout the paper, communication between populations is the Boolean spike variable s_i^P; continuous state is restricted to membrane potentials, synaptic efficacies, currents, and eligibility traces.

Implementation and reproducibility

The network was implemented with CoNeX 0.1.5 and PyMoNNtorch 0.1.4 [21–23]. Neuronal populations were

`NeuronGroup` objects and projections were `SynapseGroup` objects. The simulations used Python 3.14, PyTorch 2.9.1, CUDA 12.6, and an NVIDIA RTX 4060 GPU. Unless stated otherwise, endpoint experiments used seed 42; replication used seeds 42, 142, 242, 342, and 442. Checkpoints, cached cortical event tensors, scripts, figures, JSON reports, and SHA-256 hashes were saved with the manuscript artifacts. The reported joint checkpoints have SHA-256 hashes `8bcfe512...f8c1fca` (Caltech) and `3ed14902...aca9f80b` (MNIST).

Datasets and train/validation splits

The Caltech binary task used the first 200 lexicographically sorted images from the *Faces_easy* category and the first 200 from *Motorbikes* in Caltech-101 [24]. The 400 images were deterministically permuted and split 80:20 into 320 training and 80 validation images. MNIST [25] was split from the torchvision training partition; the endpoint used 20,000 training and 5,000 validation images. Labels were used only for decision-layer outcome modulation after a free trial. L4 feature learning, both directions of feature-location binding, preprocessing, saccade generation, and event caching were label agnostic. Tests with permuted labels confirmed that binding weights were unchanged. Shuffled-label controls repeated decision learning with permuted training labels.

Preprocessing, saccades, and retinal spike encoding

Images were resized to 80×80, converted to grayscale, min-max normalised, and multiplied by five. A zero-mean difference-of-Gaussians (DoG) filter approximated a centre-surround receptive field,

$$K(\mathbf{r}) = \frac{1}{2\pi\sigma_c^2} e^{-\|\mathbf{r}\|^2/(2\sigma_c^2)} - \frac{1}{2\pi\sigma_s^2} e^{-\|\mathbf{r}\|^2/(2\sigma_s^2)}, \quad (4)$$

with $\sigma_c = 2$ px and $\sigma_s = 8$ px. The ON response $r = \lfloor K * I \rfloor_+$ was normalised by the 99th percentile and clipped to $[0, 1]$. Nine 35×35 patches were sampled on a fixed 3×3 grid centred at rows and columns $\{20, 40, 60\}$, corresponding to the nine L5/6 reference-frame locations.

Each pixel p in a patch drove a deterministic phase encoder,

$$\phi_p(t+1) = \phi_p(t) + \rho_p, \qquad s_p^{\text{ret}}(t+1) = \mathbb{1}[\phi_p(t+1) \geq 1], \quad (5)$$

$$\phi_p(t+1) \leftarrow \phi_p(t+1) - s_p^{\text{ret}}(t+1), \quad (6)$$

where ρ_p is the normalised DoG response. Caltech used 32 sensory steps per saccade and MNIST used 48.

L4 convolutional feature maps

L4 used M convolutional feature maps ($M = 25$ Caltech, $M = 100$ MNIST), each with an 11×11 kernel $W_m^{(4)}$. Retinal spikes were convolved into feature-map current,

$$I_m^{(4)}(x, y, t) = \sum_{u,v} W_m^{(4)}(u, v)\, s^{\text{ret}}(x + u, y + v, t). \quad (7)$$

The L4 membrane followed equations (1) to (3) with $\lambda_4 = 1$ and $\theta_4 = 1$. During a sensory integration phase the membrane accumulated convolutional current; during a release phase it emitted bursts by subtractive reset, yielding integer-like spike counts through Boolean events.

Kernels were pretrained by local convolutional STDP with spatial competition [11–13]. Presynaptic and postsynaptic traces x_{pre} and x_{post} decayed with $\tau = 5$ steps. For a winning postsynaptic unit, the kernel update was

$$\Delta W_m^{(4)} = \frac{1}{N} \left(A_+ x_{\text{pre}} s_{\text{post}} - A_- s_{\text{pre}} x_{\text{post}}\right), \quad (8)$$

with $A_+ = 1.0$, $A_- = 0.1$, and N the number of kernel elements. A local three-winner spatial competition limited each update. Kernels were clipped to $[0, 1]$, normalised to total weight 60, recorded at initial, early (20 samples), middle (100 samples), and final (200 samples) stages, and then frozen before joint cortical acquisition.

L2/3 pooling, K-WTA sparsity, and apical current

Each L4 feature map was adaptive-average-pooled to a 5×5 grid and flattened, producing $F = 625$ L2/3 units for Caltech and $F = 2500$ for MNIST. The L2/3 membrane received pooled basal current and optional apical current,

$$v_f^{(3)}(t+1) = \lambda_3 v_f^{(3)}(t) + I_f^{\text{pool}}(t) + g_a \tanh\bigl(I_f^{\text{api}}(t)\bigr), \quad (9)$$

with $\lambda_3 = 1$ and apical gain $g_a = 0.01$. A k-winners-take-all operator retained the k largest membranes as spikes,

$$s_f^{(3)}(t) = \mathbb{1}\left[v_f^{(3)}(t) \in \text{TopK}\{v^{(3)}(t), k\}\right], \quad (10)$$

where $k = 5$ for Caltech and $k = 16$ for MNIST. We interpret this operator as a compact abstraction of local inhibitory competition rather than a detailed interneuron model [26]. The sparse winner code was held during the decision window for the current saccade.

The apical projection from L5/6 location ℓ to L2/3 feature f had weight $A_{\ell f}$. During joint acquisition it learned by local coactivity with mild decay and clipping [27],

$$A_{\ell f} \leftarrow \text{clip}_{[0,1]}\left((1 - \lambda_A) A_{\ell f} + \eta_A \langle s_\ell^{(56)} s_f^{(3)} \rangle\right), \quad (11)$$

with $\eta_A = 0.002$ and $\lambda_A = 0.001$. During inference, $I_f^{\text{api}} = \sum_\ell A_{\ell f} s_\ell^{(56)}$.

The reciprocal feature-to-frame projection from L2/3 feature f to L5/6 location ℓ had weight $B_{\ell f}$. It was updated concurrently from the same spike events by local pre/post coactivity,

$$B_{\ell f} \leftarrow \text{clip}_{[0,1]}\left((1 - \lambda_B) B_{\ell f} + \eta_B \langle s_\ell^{(56)} z_f^{(3)} \rangle\right), \quad (12)$$

where $z_f^{(3)}$ is an exponentially decayed L2/3 trace ($\tau_B = 8$), $\eta_B = 0.2$, and $\lambda_B = 0$. After each training epoch, rows of B were L2-normalised, implementing heterosynaptic competition across the features that project to each L5/6 location.

L5/6 current-driven reference-frame LIF layer

L5/6 contained nine LIF neurons arranged as a 3×3 object-centred reference frame. For location command $m_\ell(t) \in \{0, 1\}$ and reciprocal query gain g_r, the command and optional L2/3 cue were converted to current rather than to an assigned spike,

$$I_\ell^{(56)}(t) = g_m m_\ell(t) + g_r \sum_f B_{\ell f} s_f^{(3)}(t), \quad (13)$$

and the membrane followed

$$v_\ell^{(56)}(t+1) = \lambda_{56} v_\ell^{(56)}(t) + I_\ell^{(56)}(t), \quad (14)$$

$$s_\ell^{(56)}(t+1) = \mathbb{1}[v_\ell^{(56)}(t+1) \geq \theta_{56}], \quad (15)$$

with $\lambda_{56} = 0.9$, $g_m = 4.0$, and $\theta_{56} = 0.5$. The active neuron was reset after spiking. During ordinary object decisions $g_r = 0$, so the externally specified motor-efference sequence defines the current location without visual retrieval current. During feature-to-frame assays, $m_\ell = 0$ and $g_r = 1$, so held-out L2/3 spikes alone drive the learned reciprocal projection into the L5/6 LIF population. We did not learn the reference frame or close the perception-action loop.

Spiking decision layer and online dopamine-gated eligibility

The decision layer contained one LIF neuron per class. Its presynaptic event was the coincidence of a L2/3 feature spike and the current L5/6 location spike,

$$g_{\ell f}(t) = s_\ell^{(56)}(t)\, s_f^{(3)}(t). \tag{16}$$

Each class c had non-negative excitatory and inhibitory banks $W^+_{c\ell f}$ and $W^-_{c\ell f}$, clipped to $[0, 5]$. Decision current was

$$I_c^{(2)}(t) = I_0 + \gamma \sum_{\ell,f} g_{\ell f}(t) \left(W^+_{c\ell f} - W^-_{c\ell f} \right), \tag{17}$$

with gain $\gamma = 0.3$ and tonic current $I_0 = 0.5$. The decision membrane used $\lambda_2 = 1$ and $\theta_2 = 1$, with subtractive reset. Class evidence was the cumulative emitted spike count,

$$n_c = \sum_t s_c^{(2)}(t), \qquad \hat{y} = \arg\max_c n_c. \tag{18}$$

No linear classifier, softmax, or rate-vector readout was trained on top of these spikes.

Learning used an online three-factor rule [15–17]. During the free trial, presynaptic coincidence events and natural postsynaptic decision spikes laid down eligibility traces,

$$E_{\ell f}(t+1) = \rho_E E_{\ell f}(t) + g_{\ell f}(t), \tag{19}$$

$$H_c(t+1) = \rho_E H_c(t) + s_c^{(2)}(t), \tag{20}$$

where $\rho_E = \exp(-1/\tau_E)$, $\tau_E = 128$ steps. Before modulation, traces were converted to saturated tags

$$\bar{E}_{\ell f} = \mathbb{1}[E_{\ell f} \geq \epsilon_E], \qquad \bar{H}_c = \mathbb{1}[H_c \geq \epsilon_E], \tag{21}$$

with $\epsilon_E = 0.1$. After the free decision, the label selected a target class y and the highest-count non-target competitor j. If the target margin $n_y - n_j$ was below margin m ($m = 2$ Caltech, $m = 6$ MNIST), class-specific modulatory pulses were assigned,

$$D_c = \begin{cases} +1, & c = y, \\ -1, & c = j, \\ 0, & \text{otherwise.} \end{cases} \tag{22}$$

For sample b and class c, the eligible location-feature field was $Q_{bc\ell f} = \bar{H}_{bc} \bar{E}_{b\ell f}$ and was normalised as $\widetilde{Q}_{bc} = Q_{bc} / \max(1, \|Q_{bc}\|_2)$. This prevents a sample with more tagged synapses from receiving a larger total update solely because of tag count. The update used only this free-trial eligibility; no output spikes were replayed or forced:

$$\Delta W^+_{c\ell f} = \eta_D \sum_b [D_{bc}]_+ \, \widetilde{Q}_{bc\ell f}, \tag{23}$$

$$\Delta W^-_{c\ell f} = \eta_D \sum_b [-D_{bc}]_+ \, \widetilde{Q}_{bc\ell f}. \tag{24}$$

Here $[x]_+ = \max(x, 0)$, $\eta_D = 0.02$ for Caltech and $\eta_D = 0.05$ for MNIST.

Joint acquisition and fixation-gated plasticity

After L4 pretraining, apical, reciprocal, and decision plasticity operated within the same acquisition trials. The fixed retinal–L4–L2/3 front-end was first run without plasticity to cache deterministic binary L2/3 event sequences; this avoids recomputing the fixed convolutional front-end during repeated acquisition epochs. Each cached sequence retained the nine ordered saccades and was presented once per free trial.

At the start of fixation q, the short apical and reciprocal pre/post traces were cleared to prevent residual spikes from fixation $q-1$ being assigned to the new L5/6 location. One simulation step established the current L2/3 winner spikes and current-driven L5/6 LIF spike with binding plasticity closed. Both binding projections then opened for one coactivity step, applying equations (11) and (12). They closed for the remaining decision steps at that fixation. In contrast, decision eligibility in equation (20) was not reset between fixations and therefore accumulated over the complete object-level sequence. After the ninth saccade, the free decision was read out and the class label generated equation (22); this modulated only the decision projection.

Both binding pathways remained plastic throughout all three Caltech acquisition epochs and all eight MNIST acquisition epochs. MNIST then received two decision-only consolidation epochs with binding plasticity closed. No sensory, L2/3, L5/6, or decision spike was replayed during learning. Because cached L2/3 events are used, “online” here denotes immediate local updates within each presented cortical spike trial, not a recurrent update of the raw-sensory front-end across epochs.

Causal ablations and readout controls

For ordered, reversed, shuffled, and silent L5/6 controls, the cached L2/3 event sequence was held fixed and only the L5/6 sequence was modified. The non-spatial decision control averaged $W^{\pm}$ over the location axis before inference. The no-dopamine control evaluated an untrained decision layer with the same event stream. The apical ablation zeroed $A_{\ell f}$ during raw-image inference. Matched non-spiking readouts were fitted only as analysis baselines: a sparse logistic readout on spatial L2/3 events, a location-agnostic pooled-rate readout, and a shuffled-label spatial readout. These readouts are not part of the proposed model.

Bidirectional and prospective binding assays

For feature-to-frame retrieval, held-out L2/3 spike cues drove the learned reciprocal $B_{\ell f}$ synapse inside the fully spiking column. Motor current was clamped to zero, reciprocal gain was set to $g_r = 1$, and the L5/6 LIF population emitted location spikes for six query steps. Performance was measured as top-k recovery of the true saccade location from L5/6 spike counts; synaptic-current ranks were reported as a secondary diagnostic. Shuffled-binding controls permuted the learned association.

For frame-to-feature priming, visual and basal L2/3 input were clamped to zero. A one-hot location command was injected as current into the corresponding L5/6 LIF unit for eight settle steps, and the apical L2/3 membrane was compared with the held-out L2/3 winners observed at that location. Controls used reversed locations, shuffled apical binding, a global feature prior, and a random baseline. Average precision and recall at k quantified whether the stimulated reference-frame location primed the features that would be expected there.

Caltech Faces localization and detection

The face search experiment froze the Caltech checkpoint trained only on *Faces_easy* and Motorbikes. It then evaluated non-centered *Faces* images from Caltech-101. Candidate windows were generated over a 91-view grid; each candidate was resized to the column input size and processed by the same spiking model. The selected localization was the candidate with maximal ordered face evidence. Annotations were used only for final IoU, CorLoc (IoU ≥ 0.5), pointing accuracy, and centre-error evaluation.

Table 2. Main simulation parameters.

Component	Caltech	MNIST
Input and saccades	80×80 images; 35×35 patches; nine centres at $\{20, 40, 60\}^2$	Same
Retinal phase encoder	32 sensory steps; DoG $\sigma_c = 2$, $\sigma_s = 8$	48 sensory steps; same DoG
L4	25 maps; 11×11 kernels; $\theta_4 = 1$; STDP $A_+ = 1$, $A_- = 0.1$	100 maps; otherwise same
L2/3	5×5 pooled grid per map; $F = 625$; $k = 5$ winners	$F = 2500$; $k = 16$ winners
L5/6	Nine current-driven LIF units; $\lambda_{56} = 0.9$; $g_m = 4.0$; $\theta_{56} = 0.5$	Same
Apical L5/6→L2/3	$\eta_A = 0.002$; decay 0.001; gain $g_a = 0.01$; clip $[0, 1]$	Same
Reciprocal L2/3→L5/6	$\eta_B = 0.2$; $\tau_B = 8$; decay 0; clip $[0, 1]$; row L2 normalisation; $g_r = 0$ for ordinary decisions and $g_r = 1$ for retrieval assays	Same
Decision LIF	Two class neurons; $\gamma = 0.3$; $I_0 = 0.5$; $\theta_2 = 1$; 8 decision steps	Ten class neurons; otherwise same
Decision plasticity	$\eta_D = 0.02$; margin $m = 2$; $\tau_E = 128$; tag threshold 0.1; weight clip $[0, 5]$	$\eta_D = 0.05$; margin $m = 6$; otherwise same
Training schedule	Three joint acquisition epochs; binding and decision plasticity concurrent	Eight joint acquisition epochs plus two decision-only consolidation epochs

For detection, the rule family and operating threshold were selected on development source indices 1–60 from *Faces* and Motorbikes. Test images were source-disjoint: *Faces* indices 201–435, Motorbikes indices 201–435, 150 *BACKGROUND_Google* images, and 150 hard distractor images from seven unseen categories. The selected score was the maximum ordered evidence,

$$S = \max_q \left[(n_{\text{face},q} - n_{\text{motor},q}) + 0.49 \tanh(\Delta W_q) \right], \qquad (25)$$

where q indexes candidate windows and ΔW_q is the gated synaptic margin. The bounded synaptic term breaks ties but cannot overturn a one-spike decision margin.

Statistics

Accuracy intervals are exact Clopper-Pearson intervals [28]. Paired binary prediction differences use exact McNemar tests [29]. Mean paired differences and average-precision intervals use nonparametric bootstrap resampling [30]; the detection AP interval is stratified by positive/negative test group. Reported p values are descriptive controls for the planned perturbations rather than corrections for exploratory model selection.

Parameter summary

Data and code availability

The repository contains the training and evaluation scripts, saved checkpoints, cached cortical spike-event tensors, figure-generation code, and machine-readable JSON reports. MNIST is obtained through torchvision and Caltech-101 from its public archive. The canonical joint checkpoints are saved under `outputs/joint_training/caltech_all_acquisition_fixation_gated` and `outputs/joint_training/mnist_all_acquisition_fixation_gated`.

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